TEXORA · LECTURE SHEET

Dobby Shedding

The shedding motion between the tappet and the jacquard: a pattern chain of pegged lags selects which hooks a reciprocating knife shall lift, so that up to about two dozen heald shafts can be raised in any order the design demands.

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1. Why dobbies exist — beyond tappet capacity

In **tappet (cam) shedding** each shaft is driven by its own cam, whose profile is cut once and for all. The arrangement is simple, positive and fast, but it carries three hard limits. First, only about **8 shafts (rarely 10)** can be mounted side by side, because each cam needs width, a follower and clearance. Second, the **pick repeat is short** — fixed by the cam set, commonly 4–8 picks. Third, **changing the design means changing cams**: a new set must be cut, fitted and timed, which is slow and expensive.

Real cloth quickly outgrows those limits: a pointed twill, diamond, check or herringbone may need 12–20 shafts with a pick repeat of 16, 32 or more picks in no simple rotation, and cutting a cam set for every such design is impractical. The industry's answer is the **dobby**: the lift of every shaft on every pick is decided not by a fixed cam profile but by **pattern information that can be changed cheaply** — pegs driven into lags joined into an endless chain; later, holes in a paper tape; today, an electronic signal. Change the chain or the file, and the same machine weaves a different design.

2. The working principle in one view

A dobby separates two jobs that the tappet merges into one cam: **selection** — deciding which shafts rise on this pick — and **power** — supplying the lifting force. Power comes from one or two **knives**: heavy blades reciprocating once per pick. Selection is done by vertical **hooks**, one (or a pair) per shaft: a hook placed **over the knife** is caught and carried by it; a hook left out of its path is not moved at all.

The classical peg-and-lag dobby works as follows. The lags form an endless **pattern chain** over a rotating **pattern cylinder** (barrel), indexed by one lag per pick, so a fresh lag is presented for every pick. Where a **peg** stands in the lag, it presses a **feeler (needle)** which, through a small lever system, pushes the corresponding hook into the knife's path. Where there is **no peg** (a blank), the feeler is not pressed, the hook stays clear, and its shaft is not lifted. Lifted hooks pull on **jacks (levers)**, which draw on cords to the **heald shafts**, raising them to the top shed line. In a **negative dobby** the shaft returns by springs or its own weight; in a **positive dobby** a mechanism drives it down again (Section 4).

The same logic survives every later medium: punched holes in a **paper tape** admit the feelers; a **rotary dobby** locks or releases eccentric units on a rotating shaft; an **electronic dobby** energises electromagnets or pneumatic valves from the stored pattern. Peg, hole or signal present means "lift"; absent means "do not lift".

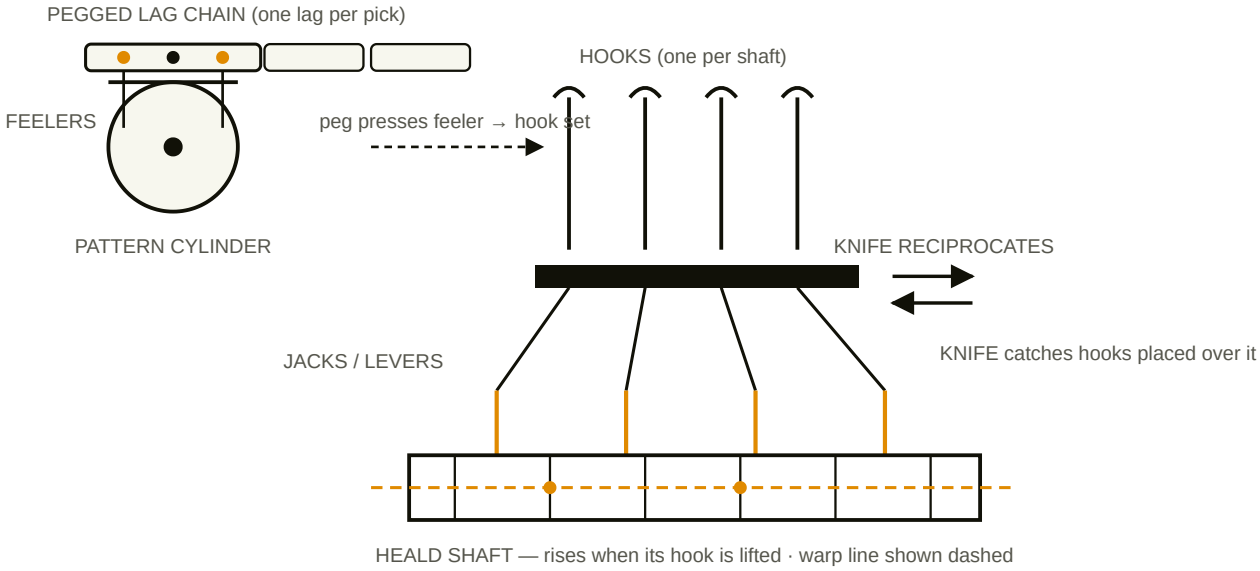


Figure 1 — Dobby schematic (peg-and-lag, simplified). Pegs set the feelers, the feelers place selected hooks over the reciprocating knife, and the lifted hooks raise their shafts through jacks and cords.

3. The working cycle, pick by pick

1. **Indexing.** The pattern cylinder turns by one lag pitch and the new lag rests over the feelers — exactly one lag per pick; a chain that jumps a lag weaves the design one pick out of step.
2. **Selection.** The feelers are brought against the lag. Where a peg stands, the feeler is pressed back and its lever swings the hook's beak over the knife; where the lag is blank, a spring keeps the hook clear.
3. **Lifting.** The knife makes its working stroke, carrying every hook resting on it; the movement passes to the jack, whose cord raises the shaft by the full shed height.
4. **Dwell and pick.** The knife pauses with the shed fully open while the pick is inserted; the dobby must not creep, or the shed closes on the pick.
5. **Return.** The knife returns; negative-dobby shafts are pulled down by their return springs, positive-dobby shafts are driven down. The cylinder indexes the next lag, and the cycle repeats.

Note: **selection must be complete before the knife starts its stroke** — a hook still moving is missed (a float in the cloth) or half-caught and strained. Timing cylinder, feelers and knife to the crank cycle is the heart of dobby setting.

4. Types of dobby

4.1 Single-lift and double-lift

A **single-lift dobby** has one knife and one set of hooks, completing the whole cycle every pick. A shaft that must stay up for consecutive picks is lowered and raised again between them, because the knife must return before it can lift again: the shed **closes and re-opens between every pair of picks**, wasting motion and limiting speed.

A **double-lift dobby** has **two knives and two hook sets** working alternately (odd picks, even picks). Each set has a two-pick interval to return and re-select, and a shaft that must stay raised is handed over between sets **without the shed closing** — smoother, kinder to the warp and faster, at the cost of double the hooks and pegging. Most modern power-loom dobbies are double-lift.

4.2 Negative and positive dobbies

In a **negative dobby** the knife lifts the selected shafts, but nothing drives them down: return is by **springs** or gravity. Simple and once universal, but the springs must close the shed at speed, loading the lifting mechanism and unsteading heavy shafts.

In a **positive dobby** the shafts are driven **both up and down** — by paired knives or a rotary drive — so no return springs are needed, the shed is held firmly in both positions, and heavier shafts and higher speeds become practical.

4.3 Pattern medium: lags, paper tape, rotary and electronic

FORM	PATTERN CARRIER	SELECTION MADE BY	NOTES
Peg-and-lag dobby	Endless chain of lags with driven-in pegs	Pegs pressing feelers	Classical form; pegging visible and easy to check
Paper-pattern dobby	Endless punched paper or film tape	Holes admitting feelers	Light and cheap for long repeats; tape is fragile, keep clean and dry
Rotary dobby	Pattern read from tape, card or memory	Locking units on a rotating shaft	Positive action, high speed; standard on modern shuttleless looms
Electronic dobby	None — pattern held in the controller	Electromagnetic or pneumatic actuators	Instant pattern change; no chain to build or store

5. Capacity — how big a pattern can a dobby weave?

A dobby's pattern capacity has two independent dimensions, and confusing them is a common examination error:

- **Shaft capacity (pattern width).** One hook set serves one shaft, so shafts equal hooks. A conservative working range is **12–24 heald shafts** — 16 and 20 are very common frame sizes. Modern electronic and rotary dobbies go higher, commonly to about 28–32 shafts.
- **Pick capacity (pattern length).** One lag serves one pick, so the pick repeat equals the number of lags in the chain. Chains of a few hundred lags are normal — far beyond any cam — and with tape or electronic control the pick repeat is effectively unlimited.

A dobby design is thus a rectangle of (shafts) ends by (lags) picks. Designs needing more independently controlled ends than there are shafts belong to the jacquard (Topic 6).

6. From design to chain: draft and pegging plan

Three documents control a dobby weave, and they must agree pick by pick:

- The **weave plan (design)** shows ends across and picks up; a mark means that end is *up* at that pick.
- The **draft** shows through which shaft each end is drawn. With a **straight draft**, end 1 goes on shaft 1, end 2 on shaft 2, and so on, repeating every (number of shafts) ends.
- The **pegging plan (lifting plan)** shows which shafts are lifted on each pick. It — and nothing else — is what the chain builder pegs: one lag per pick, one peg position per shaft, a peg wherever the plan has a mark.

The rule is: **the pegging plan is the weave plan re-written in shafts**. With a straight draft, shaft *s* carries end *s*, so each pegging row is simply that pick's weave-plan row read across the shafts. With pointed or broken drafts, convert shaft by shaft: for each pick, mark the shafts carrying the ends that are up. The worked example uses the straight draft.

> **WORKED EXAMPLE — PEGGING PLAN FOR A 2/2 TWILL**
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> Weave: 2/2 twill (two up, two down, shifting one end each pick), straight draft on 4 shafts. Weave plan (ends up): Pick 1 — ends 1, 2; Pick 2 — ends 2, 3; Pick 3 — ends 3, 4; Pick 4 — ends 4, 1. With a straight draft, end *n* is on shaft *n*, so the shafts lifted on each pick are exactly the ends that are up.
>
> Pegging plan (● = peg = shaft lifted; ○ = blank). Columns are shafts 1–4, rows are lags 1–4:
> Lag 1 (Pick 1): ● ● ○ ○ — shafts 1, 2
> Lag 2 (Pick 2): ○ ● ● ○ — shafts 2, 3
> Lag 3 (Pick 3): ○ ○ ● ● — shafts 3, 4
> Lag 4 (Pick 4): ● ○ ○ ● — shafts 4, 1

Checks: each lag carries 2 pegs, matching "2 up", and each shaft column also carries 2 pegs; the diagonal stepping one shaft per lag is the twill line's signature.
Picks in repeat = 4. Lags needed = 4 per repeat — one lag per pick.

WORKED EXAMPLE — PATTERN CHAIN LENGTH

A design repeats every 48 picks; a built chain contains 240 lags.
Step 1 — one lag serves one pick, so the chain carries 240 picks of pattern.
Step 2 — full repeats = 240 ÷ 48 = **5 full repeats**.
Step 3 — check: 5 × 48 = 240 lags, exactly the chain length, so the chain joins with no part-repeat left over. (A lag count that is not an exact multiple of the repeat breaks the pattern once per chain revolution.)

7. Dobby setting and care

- **Shed timing and size.** Set the knife stroke so the shed is fully open at weft insertion and selection is finished before the knife moves; too late traps the insertion element, too early wastes dwell and strains the warp.
- **Knife and hook condition.** Knife edges straight and clean; bent or burred hooks are the commonest cause of missed lifts, and a hook standing half-on the knife is strained at every pick.
- **Chain building and checking.** Peg strictly from the plan, then hand-turn the cylinder through one full repeat, comparing every lag with the plan — one wrong peg repeats its fault every repeat of the piece. Count lags when a chain is joined.
- **Feelers, springs and cleanliness.** Feelers must move freely; fluff around the feeler box causes missed selections that look like random faults. Clean daily; oil guides, bearings and pivots to schedule, sparingly near the chain. On negative dobbies, replace weak return springs as a set. After any repair, check all shafts hang level — an uneven shed cannot be cured by timing alone.

8. Dobby versus tappet — comparison

FEATURE	TAPPET (CAM) SHEDDING	DOBBY SHEDDING
Pattern stored as	Cam profiles — cut metal, one cam per shaft	Pegs in lags, holes in tape, or a file
Heald shafts	About 8 (rarely 10)	Commonly 12–24; electronic dobbies higher

Pick repeat	Short — fixed by the cam set, 4–8 picks	Long — one lag per pick; hundreds of lags
Design change	Slow, costly — new cams cut and timed	Quick — re-peg the chain or load a file
Typical cloths	Plain weave, simple twills, satins	Multi-shaft twills, checks, diamonds, small geometrics
Action	Positive both ways (cam and follower)	Negative or positive; single- or double-lift
Speed capability	High for simple weaves	Lower for single-lift mechanical types; high for rotary/electronic
Cost and complexity	Lowest of the three systems	Intermediate — more parts, pegging labour, setting skill

The dobby's pattern unit is the **shaft**: more shafts and picks than cams allow, at a fraction of the jacquard's cost.

Glossary

Dobby — a shedding motion in which pattern information selects which shafts a reciprocating knife lifts each pick. **Knife** — the reciprocating blade supplying the lifting power. **Hook** — the vertical element, one per shaft, placed over or clear of the knife. **Jack** — the lever transferring a hook's movement to the shaft cords. **Lag** — one bar of the pattern chain, carrying one pick of peg positions. **Pegging plan** — the chart of shafts lifted on each pick, from which the chain is pegged. **Pattern cylinder** — the barrel carrying the chain, indexed one lag per pick. **Feeler** — the finger reading each lag's pegs. **Negative dobby** — shafts return by springs or gravity. **Positive dobby** — shafts are driven both ways. **Double-lift dobby** — two alternating knife-and-hook sets, so a shaft can stay up across picks.

Key points

• The dobby exists because tappets are limited in shafts (about 8), pick repeat and cost of changing cams.
• Selection and power are separated: pegs, holes or a signal select the hooks; the knife supplies the lift.
• Peg present = hook over the knife = shaft lifted; blank = hook clear = not lifted.
• Capacity has two dimensions: shafts (commonly 12–24; electronic dobbies higher) and picks (one lag per pick).

• The pegging plan is the weave plan re-written in shafts; one lag per pick of the repeat.
• Double-lift dobbies hand a raised shaft between knife sets so the shed need not close; positive dobbies drive shafts down as well as up.

Practice

1. A design repeats every 36 picks; a chain is built with 288 lags. How many full design repeats does the chain carry? Show the steps.
2. Write the pegging plan (a 4-lag grid of peg/blank) for a 3/1 twill on 4 shafts, straight draft, and state how many pegs each lag carries.
3. A fabric needs 18 shafts and a 64-pick repeat. Explain why tappet shedding cannot weave it but a dobby can.
4. A shaft must stay raised for three consecutive picks. Explain how a double-lift dobby achieves this without the shed closing, and what a single-lift dobby does instead.
5. A dobby fabric shows one end floating wrongly at the same position in every design repeat, the rest of the pattern being correct. Give the most likely cause and the first check you would make.
6. Give two advantages and two disadvantages of electronic dobby control.

Answer key

1. Full repeats = 288 ÷ 36 = **8 full repeats** (one lag per pick). Check: 8 × 36 = 288 lags.
2. 3/1 twill: three ends up per pick, shifting one end each pick — Pick 1: shafts 1, 2, 3; Pick 2: 2, 3, 4; Pick 3: 3, 4, 1; Pick 4: 4, 1, 2. Grid (● = peg): Lag 1: ● ● ● ○; Lag 2: ○ ● ● ●; Lag 3: ● ○ ● ●; Lag 4: ● ● ○ ●. Each lag carries **3 pegs**, matching "3 up".
3. Tappet: only about 8–10 cams can be fitted, and a cam set's pick repeat (4–8 picks) is far below 64. Dobby: 18 shafts is within the 12–24 capacity, and 64 picks need only 64 lags.
4. The two sets work alternately (odd picks, even picks). The first set lifts the shaft; before it must return, the second set — its hook also selected — takes over, so the shaft is held up continuously and the shed never closes. A single-lift dobby's set must return after every pick, lowering and re-raising the shaft each time.
5. Most likely a single peg wrongly placed (or missing) in one lag — a fault once per repeat, at the same pick and shaft, is the signature of a pegging error. First check: compare that lag against the pegging plan.
6. Advantages (any two): instant file-load pattern change; no pegging errors or chain wear. Disadvantages (any two): higher capital cost; dependence on electronics support.